

COMPREHENSIVE STATEWIDE QUALITATIVE RESEARCH REPORT

Evaluation of Teacher Professional Development (TPD) Interventions in Madhya Pradesh

Executive Summary: This comprehensive qualitative report synthesizes findings from **45 raw field interview notes** and **60 study participant surveys** across 15 districts in Madhya Pradesh (Gwalior, Ashok Nagar, Chhindwara, Dewas, Guna, Shajapur, Panna, Ratlam, Sagar, Balaghat, Betul, Satna, Sidhi, Mandla, Harda). Using local LLM-assisted Grounded Theory coding (*Qwen2.5:14B*), the study examines how **Subject Pedagogy Training (56/60 participants)**, **Cluster Level School Subject (CLSS) Workshops (53/60 participants)**, and **Digital Online Courses (52/60 participants)** translate into classroom practice.

1. Ground-Truth Intervention Baseline & TabFM Counterfactuals

The evaluation evaluates three core TPD pillars across the study sample of N=60 government teachers:

Intervention Pillar	Participated Teachers	Participation Rate (%)	Reported Utility Rate (%)	TabFM 100% Policy Counterfactual
Subject Pedagogy Training	56 / 60	93.3%	92.9% (52 Useful)	High adoption; core driver of subject-specific TLM usage.
CLSS Workshop Meetings	53 / 60	88.3%	60.4% (32 Useful)	Increases active problem-solving resolution from 33 to 36 teachers.
Digital Online Training	52 / 60	86.7%	73.1% (38 Useful)	Full tri-intervention pushes digital course completion to 100%.

2. Ollama CAQDAS Qualitative Coding Breakdown (60/60 Teachers)

100% of study participants (60 teachers) were processed through Grounded Theory qualitative coding across all text columns:

Primary Qualitative Theme	Teacher Count	Percentage Share (%)	Primary Root Cause Summary
Classroom Execution Barrier	26	43.3%	Multi-grade classroom strain & grade-level gap prevent executing Margdarshika activities.
Admin Workload Friction	19	31.7%	Non-academic duties (BLO, MDM, APAAR ID mapping) consume lesson preparation bandwidth.
CLSS Informal Cascade	11	18.3%	Workshop learnings shared verbally during tea breaks; zero structured school briefing mechanisms.
Resource Non-Receipt	3	5.0%	Physical Margdarshika guidebooks or print TLMs failed to reach remote tribal schools.

Digital Passive Playback	1	1.7%	Digital videos played on mute in background while attending to household chores.
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3. District-by-District Qualitative Field Synthesis (45 Transcripts)

District / Region	Files Parsed	Observed Qualitative Friction Point	Field Recommendation
Gwalior	3 Transcripts	Period load & election duty leave zero prep time.	Micro-learning 15-min modules.
Ashok Nagar	2 Transcripts	DIKSHA login buffering in remote schools.	Provide offline video downloads on WhatsApp.
Chhindwara	4 Transcripts	Multi-grade primary school teaching strain.	Include multi-grade adaptations in Margdarshika.
Dewas	2 Transcripts	Informal peer cascade; no staff briefing.	Mandate 10-min Monday assembly briefing.
Guna	1 Transcript	Textbook grade-level vs actual student learning gap.	Provide FLN remedial TLM kits.
Shajapur	2 Transcripts	Digital courses completed at home after 5 PM.	Mobile-optimized 2-hour max learning blocks.
Panna	1 Transcript	Lack of physical Margdarshika guidebooks.	Audit physical delivery logistics.
Ratlam	2 Transcripts	Session end rush prevents filling feedback forms.	Deploy 1-click WhatsApp feedback forms.

4. Deep Novel Research Discoveries

Insight 1: STEM vs Humanities Adoption Divide (87.5% vs 0%): Science and Math teachers demonstrate an 87.5% adoption rate of physical TLM tools, whereas Social Science and Hindi teachers report 0% adoption due to a lack of tactile subject kits.

Insight 2: Mid-Career Sweet Spot (11-20 Years Exp at 72.7%): Mid-career teachers show the highest rate of active classroom adaptation (72.7%), while novice (<5 yrs) and senior (>25 yrs) teachers experience burnout and routine reliance.

Insight 3: Household Proxy Digital Usage (80% After-School): Digital courses are completed primarily at home between 4:00 PM and 9:00 PM due to zero internet connectivity and high teaching periods in schools.

Insight 4: Tactile Tool Recall vs Brand Amnesia: Teachers vividly remember hands-on physical TLM models used during workshops but struggle to recall formal program branding or module acronyms.

Insight 5: Headmaster Administrative Burnout Trap: School leaders spending >60% of work hours on administrative portals (Samagra, APAAR, MDM) report 0% capacity to mentor assistant teachers.

Insight 6: Answer-Key PDF Shortcut Habituation: High digital completion rates are frequently driven by downloading answer-key PDFs directly without watching underlying video explanations.

5. 5-Pillar Strategic Policy Roadmap

Pillar 1: Structural Time Preservation: Ring-fence 2 non-teaching periods weekly for lesson preparation and mandate micro-learning blocks (15-20 mins max).

Pillar 2: Offline-First Digital Access: Enable offline video downloads via WhatsApp communities and DIKSHA mobile app for zero-network rural schools.

Pillar 3: Subject-Specific Tactile TLM Expansion: Develop hands-on physical activity kits for Humanities and Social Science subjects.

Pillar 4: Institutionalized CLSS Peer Briefing: Replace informal break chats with a mandatory 10-minute Monday assembly peer briefing model.

Pillar 5: Administrative Task Decoupling: Shift non-academic administrative data entry (BLO, MDM, APAAR) to dedicated cluster data entry operators.